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Mid-Term answer sheet

Q1. Overfitting = model memorises training data incl. noise -> good train acc, bad test acc.
Fixes: early stopping (stop when val loss rises), dropout (randomly drop neurons so it can't rely on a few).

Q2. $x \rightarrow W_q, W_k, W_v \rightarrow q, k, v$.
score = $q \cdot k$, softmax -> weights that sum to 1.
output = sum of weights * v.
divide by sqrt(dk) because big dot products make softmax peaky -> tiny gradients.

Q3. $P = 40/50 = 0.8$  $R = 40/60 = 0.67$
$F1 = 2PR/(P+R) = 0.73$  $Acc = 70/100 = 0.7$

(continued)

Q4. GAN - G and D compete (minimax). Diffusion - add gaussian noise step by step, train a net to predict the noise.
GAN unstable, mode collapse. Diffusion stable + diverse but slow (many steps); GAN is 1 step.

Q5. RAG: split docs into chunks -> embeddings -> vector DB. Query -> embed -> top k chunks -> into prompt -> LLM answers.
Fail 1: retrieves irrelevant chunks -> hybrid search + reranker.
Fail 2: LLM makes things up -> tell it to answer only from context and cite.